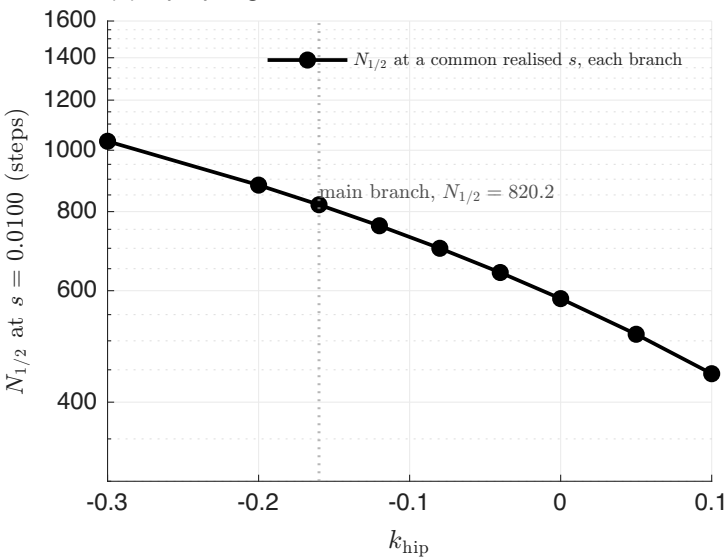
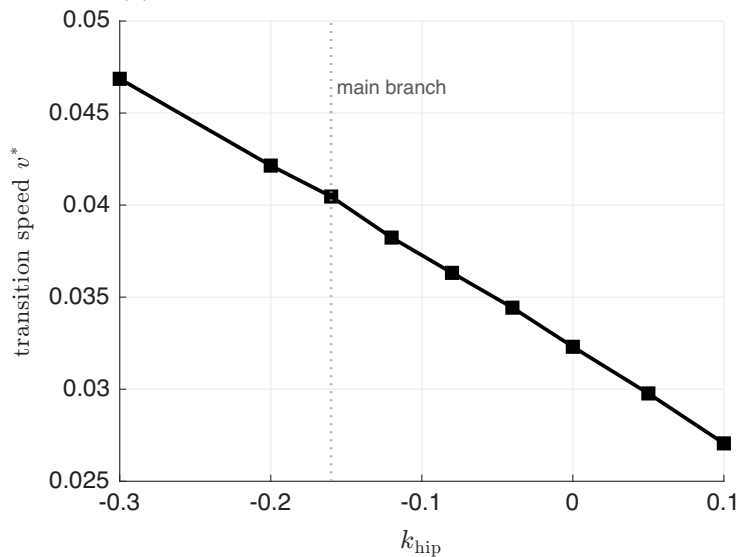


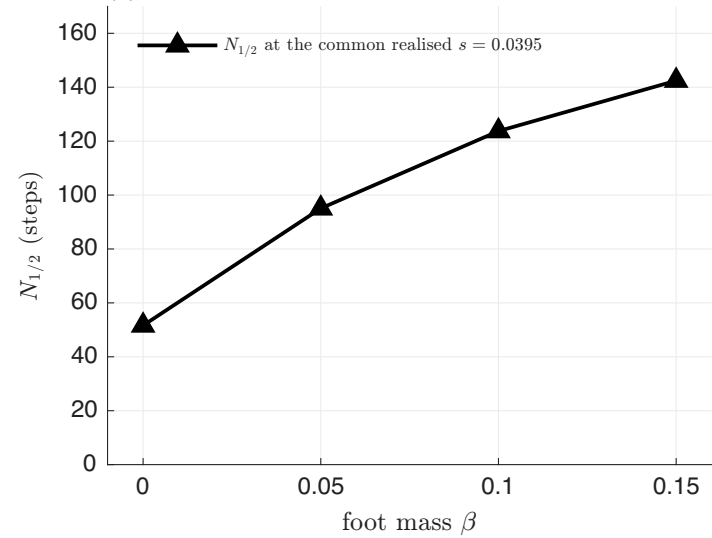
(a) hip-spring coefficient



(b) where the mode switch occurs



(c) small finite foot mass

(d) push-off coefficient in $P(q) = c \alpha_p \tan \alpha_p$ 